Vamsi Krishna

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.28
Narayana Junior college	Class XII (Telangana State Board of Intermediate Education)	2020	92.70%
VISWABHARATI HIGH SCHOOL	Class X (APSSC)	2018	10.00%

PROJECTS

Delta Hedging & Stock Path Generation Simulator [github]

- Built a quantitative finance simulator generating **1,000 synthetic GBM stock paths (375K+ minute-level price points)**, calibrated on real NIFTY option minute data.
- Computed implied volatility (bisection solver) and Black-Scholes Greeks (delta, gamma, theta) across the strike chain.
- Developed a dynamic delta hedging engine with **374 discrete rebalancing intervals**, transaction costs, portfolio valuation and final P&L computation.
- Trained an **RNN deep hedger with a CVaR loss for 100 epochs** and benchmarked it against Black-Scholes delta hedging on VaR, CVaR and P&L variance.

End-to-End Android Malware Detection System [github]

- Built a **Python static analysis & ML system** with Androguard to **automate APK feature extraction** (manifest, certificate, DEX, VirusTotal).
- Engineered scalable **data pipelines for 65K+ apps**, enabling automated malware classification.
- Optimized ML models (Random Forest, XGBoost, SVM) achieving **F1 = 0.97, ROC-AUC = 0.994**

DNS Security Validator -- DNSSEC Chain-of-Trust Auditor [github]

- Built a Python **DNSSEC auditor** from **RFC 4033-4035/5155**: **RRSIG, DNSKEY, and DS** chain validation and **NSEC/NSEC3** proofs.
- Engineered a **root-anchored resolver** (**KSK-2017/2024**) classifying every zone cut as **SECURE / INSECURE / BOGUS**.
- Simulated **6 tampering attacks** (spoofing, forged keys, downgrade) with **100% detection** across **3 validation stages**.
- Shipped a **web dashboard** (**0-100 scoring**, **~2 s audits**) with **67 unit tests** and a **4-container BIND lab**.

PDKind -- Model-Checking Engine (C++, Pono SMT Framework) [github]

- Implemented a safety-verification engine in **C++** by subclassing **Pono's SafetyProver**, overriding its core virtual methods and integrating directly with the underlying **SMT-solver API**
- Diagnosed and fixed a **critical soundness bug** via a self-built evaluation harness -- a probe-logic error was producing **403/403** trivially-true results -- by rewriting the core generalization behind a proper soundness gate
- Benchmarked against two existing engines on a **32-case** suite, solving **3 cases** where the standard method timed out, with **zero correctness** disagreements

TransactiWar : Shield App -- Secure Full-Stack Banking Website [github]

- Built a full-stack PHP/MySQL banking app with account creation, deposits, and withdrawals.
- Implemented multi-layered **defenses against SQLi, XSS, CSRF, LFI, and RFI** with custom security controls.
- Designed secure **session management and input validation** to ensure end-to-end transaction integrity.
- Performed **penetration testing** on peer apps and containerized the project using Docker for consistent, secure deployment.

SKILLS

Programming: C/C++, Python, Java, Bash, SQL.

Networking: TCP/UDP socket programming, DNS, NAT, subnetting, IPv4 & IPv6, routing, RDT & congestion control.

Security & Penetration Testing: Wireshark (packet capture), tcpdump, Scapy, Nmap, sqlmap, Nikto, Hydra, TLS/IPsec.

Databases: PostgreSQL, MySQL, Firebase.

Machine Learning : Data Preprocessing, PyTorch, scikit-learn, Matplotlib, pandas, NumPy, GANs (Wasserstein GAN with Gradient Penalty), Supervised Learning (XGBoost).

Tools & DevOps: Git, GitHub, Linux, VMware/VirtualBox, Docker.

POSITIONS OF RESPONSIBILITY

- Teaching Assistant for **Advanced Computer Networks** at IIT Hyderabad - Mentored **76** students
- Teaching Assistant for **ITP & DSA** at IIT Hyderabad - Mentored **150+** Students

CERTIFICATIONS

- Certificate of Excellence - **Indian CP Summer Camp** -- Advanced DSA & competitive programming - AlgoZenith
- Certificate of Completion - Learn Ethical Hacking A-Z: Beginner to Expert -- Networking and Security - Udemy

RELEVANT COURSES

Advanced Data Structures and Algorithms | Advanced Computer Networks | Cryptology | Foundations of Machine Learning | Network Security | Fraud Analytics using Predictive and Social Network Techniques | Topics in computing | Practical Malware Analysis | Software Verification

PUBLICATIONS

Grocery Purchase through Blockchain System -- *IEEE PuneCon 2024* · DOI

- Co-authored a **blockchain + IoT framework** for grocery supply chains: **Java-based ledger**, smart-contract-driven payment release, and real-time perishable-goods monitoring.